Short-Horizon Mean-Reversion Signals for BTCUSDT Perpetual Futures

Quantitative Research Report

Tick-to-OHLCV Pipeline · 8-Signal Composite · Cost-Aware Vectorized Backtest

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1. Introduction

This report documents an end-to-end quantitative research workflow for developing short-horizon directional signals on Binance BTCUSDT perpetual futures. The study targets a **5–15 minute holding period**, a regime that sits at the boundary between market-microstructure phenomena and the slower price discovery captured by conventional technical strategies.

Starting from two days of raw, nanosecond-resolution trade-level data (approximately 2.3 million events), the work proceeds through four stages:

1. **Data exploration and OHLCV aggregation** — data-quality audits, distributional characterisation of 1-minute returns, autocorrelation analysis, order-flow decomposition, and intraday seasonality profiling.
2. **Signal design** — eight individual signals spanning mean-reversion, order-flow, and momentum families, combined into a weighted composite with exponential smoothing and hysteresis-based position discretisation.
3. **Backtesting** — a vectorized simulation framework with realistic execution assumptions, transaction-cost sensitivity sweeps, rebalance-frequency analysis, and a walk-forward out-of-sample validation.
4. **Critical analysis** — an honest evaluation of what the results do and do not prove, including identifiable biases and the conditions required for live profitability.

The codebase is organised as a Python package (`src/`) with three companion Jupyter notebooks:

- `notebooks/01_data_exploration.ipynb`
- `notebooks/02_signal_research.ipynb`
- `notebooks/03_backtest.ipynb`

2. Data

2.1 Source and Schema

The dataset comprises two Parquet files covering UTC days **18 April 2025** and **19 April 2025**. Each row records a single aggressed trade on Binance BTCUSDT perpetual futures:

Field	Type	Description
`trade_time`	int64 (ns)	Exchange-assigned nanosecond timestamp
`side`	{BID, ASK}	BID = buyer-initiated; ASK = seller-initiated
`price`	float64	Execution price (USDT)
`quantity`	float64	Traded quantity (BTC)

Two derived columns are added during loading: `signed_qty` (positive for BID, negative for ASK) and `notional` (= price × quantity).

2.2 Coverage and Quality

Statistic	Value
Total trade records	2,308,624
Price range	\$84,260 – \$85,600
Missing values (any field)	0
Minutes with zero trades	0 / 2,880
Mean trades per minute	802
Max trades per minute	14,663
BID trades	1,144,581 (49.6%)
ASK trades	1,164,043 (50.4%)
Median trade size (BTC)	0.005

The data are *dense* (no gaps), clean (zero nulls), and well-balanced between buyer- and seller-initiated flow. Duplicate nanosecond timestamps (1,504,842 rows share a timestamp with at least one other row) reflect batched matching-engine outputs at the same microsecond, which is normal for a high-throughput exchange.

2.3 OHLCV Aggregation

Tick data are resampled into **1-minute bars** using a standard open/high/low/close/volume scheme, extended with microstructure fields:

$$\mathrm{VWAP}_t = \frac{\sum_{i \in t} p_i q_i}{\sum_{i \in t} q_i}, \qquad \mathrm{buy_ratio}_t = \frac{V_t^{\mathrm{buy}}}{V_t^{\mathrm{buy}} + V_t^{\mathrm{sell}}},$$

where V_t^{buy} and V_t^{sell} are the buyer- and seller-initiated volumes in bar t. In addition:

- **log returns**: $r_t = \ln(C_t/C_{t-1})$
- **spread**: $h_t - l_t$ (high minus low within the bar)
- **trade count**: number of individual fills per bar

A total of **2,880 one-minute bars** are produced across the two-day window.

2.4 Return Distribution

One-minute log returns exhibit the distributional properties typical of liquid crypto markets:

Statistic	Value
Mean	≈ 0
Std	≈ 0.00015 (1.5 bps)
Skewness	small, close to zero
Excess kurtosis	> 3 (heavy tails)

The QQ plot against the normal distribution confirms pronounced heavy tails — the distribution is leptokurtic. This motivates signal normalization via rolling z-scores (rather than raw levels) and robust clipping to $[-3, 3]$ to limit the influence of outlier bars.

2.5 Autocorrelation: Motivation for Mean Reversion

A critical finding is the **negative lag-1 autocorrelation** of 1-minute log returns:

Lag (bars)	Autocorrelation
1	−0.05 to −0.08
2	small, mixed
3–5	near zero

A down move at the 1-minute bar level is slightly more likely to be followed by an up move than another down move. This pattern is consistent with *bid-ask bounce* and *microstructure mean reversion*: market makers systematically quote on both sides and tighten spreads after one-sided flow. It is the primary empirical motivation for the mean-reversion signal suite developed in Section 3.

2.6 Intraday Patterns

Hourly breakdowns of volume, volatility, and trade count reveal:

- Elevated activity in the Asian morning session (roughly 00:00–04:00 UTC) and the US afternoon session (roughly 13:00–17:00 UTC).
- Return volatility is positively correlated with trade count (more activity = wider intrabar ranges).
- Buy ratio oscillates around 0.50 with no persistent directional bias, confirming the absence of a multi-hour trending regime in this sample.

3. Signal Design

3.1 Design Philosophy

All eight signals are designed around a single empirical observation: at horizons of 1–15 minutes, BTCUSDT returns exhibit *mean reversion* rather than momentum. Seven of the eight signals generate a **positive (long) reading when price or flow is depressed relative to a recent reference level**, and vice versa. The eighth signal (trade intensity) captures abnormal activity and pairs it with the contemporaneous price direction.

Signal convention throughout:

$$s_t > 0 \implies \text{go long}; \quad s_t < 0 \implies \text{go short}.$$

3.2 Individual Signals

Signal 1 — Z-Score Mean Reversion.

$$z_t = \frac{C_t - \bar{C}_{t,L}}{\sigma_{C,t,L}}, \qquad s_t^{(1)} = -z_t,$$

where $\bar{C}_{t,L}$ and $\sigma_{C,t,L}$ are the rolling mean and standard deviation of close prices over a lookback of $L = 20$ bars. A below-average price (negative z) generates a positive signal.

Signal 2 — Bollinger Band Mean Reversion.

$$s_t^{(2)} = -\operatorname{clip}\left(\frac{C_t - \bar{C}_{t,L}}{k \cdot \sigma_{C,t,L}},\ -1,\ 1\right), \quad L = 20,\ k = 2.$$

This normalizes the position within the Bollinger envelope to $[-1, 1]$ and inverts for mean reversion. Signals 1 and 2 are algebraically very similar when the clip is not binding; their correlation in practice approaches unity (see Section 6).

Signal 3 — VWAP Reversion.

$$\mathrm{VWAP}_t^{(L)} = \frac{\sum_{\tau=t-L+1}^{t} N_\tau}{\sum_{\tau=t-L+1}^{t} V_\tau}, \qquad s_t^{(3)} = -\,\mathrm{clip}\left(\frac{(C_t - \mathrm{VWAP}_t^{(L)})/\mathrm{VWAP}_t^{(L)}}{\sigma_{\mathrm{dev},t,L}},\ -3,\ 3\right), \quad L = 15.$$

Here N_τ is notional and V_τ is volume. VWAP acts as a *volume-weighted fair value* anchor, which is less susceptible to isolated large price moves than a simple moving average.

Signal 4 — Order Flow Imbalance (OFI).

$$\phi_t = \frac{V_t^{\mathrm{buy}} - V_t^{\mathrm{sell}}}{V_t}, \qquad s_t^{(4)} = -\,\mathrm{clip}\left(\frac{\bar{\phi}_t^{\mathrm{fast}} - \bar{\phi}_t^{\mathrm{slow}}}{\sigma_{\Delta\phi,t}},\ -3,\ 3\right), \quad f = 5,\ s = 20.$$

The fast-minus-slow decomposition isolates *recent* imbalance above the baseline. Excess buying ($\phi > 0$) is faded contrarily: informed order flow pushes price in the short run, but the resulting overextension mean-reverts.

Signal 5 — Trade Intensity.

$$I_t = \frac{n_t - \bar{n}_{t,L}}{\sigma_{n,t,L}}, \qquad s_t^{(5)} = \mathrm{clip}(I_t \cdot \mathrm{sgn}(r_t),\ -3,\ 3), \quad L = 20.$$

This combines an activity z-score with the sign of the contemporaneous return: a volume spike accompanied by an up move is treated as a *long* signal (informed aggressive buyers). Unlike the other signals, this is a **momentum** signal, not contrarian.

Signal 6 — RSI Mean Reversion.

$$\mathrm{RSI}_t = 100 - \frac{100}{1 + \mathrm{RS}_t}, \quad \mathrm{RS}_t = \frac{\overline{\Delta^+}_{t,P}}{\overline{\Delta^-}_{t,P}}, \quad s_t^{(6)} = \mathrm{clip}\left(-\frac{\mathrm{RSI}_t - 50}{20},\ -1.5,\ 1.5\right), \quad P = 14.$$

Gains and losses are smoothed with an exponential weighted moving average (span $= P$). The signal is centred at 50 (neutral) and maps overbought (RSI > 70) to -1 and oversold (RSI < 30) to $+1$.

Signal 7 — MA Crossover (Contrarian).

$$s_t^{(7)} = -\,\mathrm{clip}\left(\frac{\mathrm{EMA}_t^f - \mathrm{EMA}_t^s}{\sigma_{C,t,s}},\ -3,\ 3\right), \quad f = 5,\ s = 15.$$

Short-term momentum (fast EMA above slow EMA) is faded, consistent with the mean-reverting microstructure.

Signal 8 — Volume-Weighted Momentum (Contrarian).

$$m_t = r_t \cdot \frac{V_t}{\bar{V}_{t,L}}, \qquad s_t^{(8)} = -\,\mathrm{clip}\left(\frac{\sum_{\tau=t-L+1}^{t} m_\tau}{\sigma_{\Sigma m,t,2L}},\ -3,\ 3\right), \quad L = 10.$$

High-conviction up moves (large volume + positive return, cumulated over L bars) are faded.

3.3 Composite Signal and Position Sizing

All eight signals are combined into a weighted composite:

$$C_t^{\text{raw}} = \sum_{k=1}^{8} w_k s_t^{(k)},$$

with hand-specified weights that overweight the VWAP reversion signal on account of its grounding in volume-fair-value:

Signal	**Weight** w_k
Z-score MR	0.15
Bollinger MR	0.15
VWAP Reversion	0.20
Order Flow Imb.	0.15
Trade Intensity	0.05
RSI MR	0.10
MA Crossover	0.10
Volume Momentum	0.10
Total	1.00

The composite is then smoothed with a 5-bar EMA to reduce whipsaw:

$$C_t = \text{EMA}_5(C_t^{\text{raw}})\,.$$

Hysteresis Position Sizing. Continuous signals are discretised into $\{-1, 0, +1\}$ via a hysteresis rule with entry threshold $\theta_{\text{in}} = 0.15$ and exit threshold $\theta_{\text{out}} = 0.05$:

$$\text{pos}_t = \begin{cases} +1 & \text{if } \text{pos}_{t-1} = 0 \text{ and } C_t > \theta_{\text{in}} \\ -1 & \text{if } \text{pos}_{t-1} = 0 \text{ and } C_t < -\theta_{\text{in}} \\ 0 & \text{if } \text{pos}_{t-1} = +1 \text{ and } C_t < \theta_{\text{out}} \\ 0 & \text{if } \text{pos}_{t-1} = -1 \text{ and } C_t > -\theta_{\text{out}} \\ \text{pos}_{t-1} & \text{otherwise} \end{cases}$$

This *gap* between entry and exit thresholds prevents the strategy from flipping back and forth on noise. In the data, this produces an average holding period of approximately **14.7 bars (≈15 minutes)**, which sits at the upper edge of the target regime.

4. Backtesting Framework

4.1 Execution Assumptions

- **Entry price**: close of the bar *following* signal generation (one-bar delay via `signal.shift(1)`), eliminating look-ahead bias.
- **Return capture**: the strategy earns the close-to-close return of the entry bar.
- **Transaction cost model**: $\text{cost}_t = |\Delta\text{pos}_t| \times c$, where c is the one-way cost in basis points.
- **No leverage constraints** or margin calls are modeled.
- **No market impact** or queue priority effects.

4.2 PnL Decomposition

Bar-level gross PnL:

$$\pi_t^{\text{gross}} = \text{pos}_t \times r_t^{\text{fwd}},$$

where r_t^{fwd} is the forward close-to-close return.

Net PnL:

$$\pi_t^{\text{net}} = \pi_t^{\text{gross}} - |\Delta\text{pos}_t| \times \frac{c}{10{,}000}.$$

4.3 Walk-Forward Validation

To test out-of-sample generalization, a rolling walk-forward scheme is applied: a **12-hour training window** (720 bars) is used to warm up the rolling signal statistics, followed by a **4-hour test window** (240 bars) on which PnL is recorded. The window rolls forward by one test period and repeats.

Note: signal *parameters* (lookback lengths, weights) are not re-estimated in each training fold — they remain fixed. The training window serves purely to initialise the rolling statistics (means, standard deviations) that the signals depend on. This is discussed critically in Section 6.

5. Results

5.1 Information Coefficient Analysis

The Spearman rank correlation (IC) between each signal and h-bar forward returns is computed across all bars:

Signal	**1-min IC**	**5-min IC**	**10-min IC**	**15-min IC**
Z-score MR	0.046	0.097	0.092	0.085
Bollinger MR	0.046	0.097	0.092	0.085
VWAP Reversion	0.040	0.097	0.085	0.077
OFI	0.052	0.107	0.092	0.074
Trade Intensity	0.011	0.011	0.002	−0.018
RSI MR	0.056	0.116	0.110	0.102
MA Crossover	0.065	0.104	0.106	0.104
Volume Momentum	0.043	0.078	0.108	0.077
Composite	**0.063**	**0.105**	**0.099**	**0.079**

Key observations:

- All signals except Trade Intensity show positive IC across the 1–15 min range.
- IC peaks around the 5–10 minute horizon, consistent with the targeted holding period.
- Trade Intensity turns *negative* at 15 minutes — a warning sign discussed in Section 6.
- RSI and MA Crossover are the strongest individual signals at all horizons.
- The composite slightly outperforms the best individual signals at the 1-minute horizon, reflecting the benefit of diversification across signal families.

5.2 Quintile Decomposition

Sorting bars into quintiles by signal strength and measuring the mean 5-minute forward return in each bin confirms a broadly monotonic relationship for all price-based signals. Bins 1 (most

negative signal) and 5 (most positive) show opposite-signed average returns in the range of ±0.5–1 bps. Trade Intensity shows weak or reversed monotonicity, consistent with its low IC.

5.3 Gross vs Net Performance: Continuous Composite

Signal	Gross PnL	Gross SR	Net PnL (2 bps)	Net SR (2 bps)	Turnover	Win%
Z-score MR	0.031	34.9	−0.111	−118.9	708	51.9
Bollinger MR	0.027	40.4	−0.098	−143.1	623	51.9
VWAP MR	0.034	39.7	−0.130	−138.5	820	52.2
OFI	0.045	56.1	−0.152	−178.0	982	50.4
Trade Intensity	0.001	1.4	−0.424	−505.4	2122	51.3
RSI MR	0.036	49.1	−0.117	−150.3	767	52.3
MA Crossover	0.026	39.8	−0.049	−74.9	373	52.9
Vol. Momentum	0.026	32.5	−0.096	−113.5	609	51.4
Composite	**0.030**	**43.2**	−0.044	−63.2	371	51.8

Annualized Sharpe ratios assume 1-minute bars: $\sqrt{1440 \times 365}$ scaling. All strategies go to zero or negative net PnL at the representative 2 bps taker cost. The composite is the most *cost-efficient* signal: it has the *lowest turnover* among all strategies (371), a direct result of the EMA smoothing that filters out high-frequency signal reversals.

5.4 Transaction Cost Sensitivity: Composite Signal

Cost (bps)	Total PnL	Annualized Sharpe
0.00	+0.030	+43.2
0.25	+0.021	+30.3
0.50	+0.012	+17.5
0.75	+0.003	+4.7
1.00	−0.006	−8.0
1.50	−0.024	−34.3
2.00	−0.044	−63.2
3.00	−0.083	−120.6
5.00	−0.162	−235.3

The **breakeven transaction cost** is approximately **0.75 bps round-trip** for the continuous composite signal. This is a demanding threshold: it requires limit-order (maker) execution with minimal adverse selection, accessible only on exchange VIP fee tiers.

5.5 Thresholded (Discrete) Position Strategy

Using the hysteresis discretization from Section 3.3:

Cost (bps)	Total PnL	Annualized Sharpe	Profit Factor	Win%
0.00	+0.033	+33.1	1.146	52.5
0.25	+0.023	+23.3	1.099	48.8
0.50	+0.014	+13.5	1.056	48.4
0.75	+0.004	+3.8	1.015	48.1
1.00	−0.006	−5.9	0.977	47.8
2.00	−0.045	−42.8	0.847	47.0

The thresholded strategy produces **351 position changes** (175 round trips) over 2 days with a mean holding period of 14.7 bars, which falls squarely in the 5–15 minute target window. Its lower gross Sharpe (33 vs 43) relative to the continuous strategy reflects the information lost in

discretization, but its *total turnover* (391 vs 371) is similar, since the hysteresis suppresses trivial trades.

5.6 Rebalance Frequency Sweep

Holding the composite signal fixed and varying the rebalance interval Δ (how often the position is updated):

	Annualized Sharpe by Cost (bps)			
Rebal (min)	**0**	**0.5**	**1.0**	**2.0**
1	43.2	17.5	−8.0	−63.2
3	35.1	18.4	+1.8	−43.8
5	28.7	16.2	+3.6	−33.3
7	24.1	13.5	+2.9	−26.5
10	18.4	9.2	+0.0	−19.1
15	9.6	2.7	−4.1	−16.8

Gross Sharpe declines monotonically with rebalance interval: the signal edge is concentrated at the 1–3 minute horizon and has largely decayed by 15 minutes. For a cost of 0.5 bps, the peak risk-adjusted return is achieved at a 3-minute rebalance, not the longer intervals one might naively expect from a "5–15 minute strategy".

5.7 Walk-Forward Out-of-Sample Results

Metric	**Gross**	**0.5 bps**
Total PnL	+0.025	+0.011
Annualized Sharpe	+47.2	+20.1
Max drawdown	−0.004	−0.004
Win rate	51.4%	41.8%
Profit factor	1.25	1.10
Bars covered (OOS only)	2,160	2,160

The walk-forward Sharpe (47.2 gross) slightly *exceeds* the in-sample result (43.2), which may seem surprising. The most likely explanation is that the walk-forward drops the first 720 bars of warm-up period (where signals are noisier) from the PnL computation, leaving a cleaner signal in the test folds.

6. Critical Discussion

6.1 Signal Redundancy: Z-Score and Bollinger

Signals 1 and 2 are essentially the same transformation. Both compute $(C_t - \bar{C})/\sigma_C$; the Bollinger version merely clips the result to $[-1, 1]$ and applies a fixed scaling by $k = 2$. Their empirical IC values are identical to three decimal places (0.0463 vs 0.0462 at the 1-min horizon). In a production signal suite, these would be merged into a single entry and the freed weight reallocated to a genuinely orthogonal source of alpha — for example, funding-rate deviation, open-interest change, or a cross-instrument (ETH) signal.

6.2 Trade Intensity is Mis-Specified

Signal 5 is designed as a momentum signal (high volume + up move = long) inside a predominantly contrarian framework. Its IC at the 15-minute horizon is **negative** (−0.018): the strategy

would be better served by *fading* high-volume moves, consistent with the other signals. More fundamentally, trade intensity measured from 1-minute bars captures aggregate activity rather than the order-by-order dynamics that make activity measures genuinely informative. A tick-level measure (e.g. consecutive same-side trades, or the Lee-Ready-adjusted trade count) would be more appropriate.

6.3 The Walk-Forward Does Not Test Parameter Stability

The walk-forward scheme trains over a 12-hour window and tests over a 4-hour window, but *signal parameters are never re-estimated.* The 20-bar lookback for z-score, the 0.15 entry threshold for hysteresis, and the signal weights are identical across all folds. The walk-forward therefore tests *temporal stability of performance under fixed parameters*, not the ability to generalize a fitting procedure. A proper walk-forward would optimize the parameters on each training window and evaluate them on the subsequent test window, which would require a much larger dataset.

6.4 The Sharpe Numbers are Overstated

Annualized Sharpe ratios in the range of 30–50 should not be taken at face value. Several effects combine to inflate them:

1. **One-bar delay is insufficient for HFT mean reversion.** At the 1-minute bar level, a mean-reversion strategy needs to submit limit orders *within* the bar that experiences the dislocation, not at the close of the next bar. The one-bar-later entry captures only the portion of the reversion that extends into subsequent bars.
2. **Zero market impact.** Any realistically sized position in BTCUSDT perps would move the price, particularly during low-volume periods. The backtest treats every fill as if it were executed at the mid with zero price impact.
3. **Two days of data.** The statistical uncertainty on a Sharpe ratio estimated from 2,880 bars is enormous. A 95% confidence interval for a true Sharpe of 2 (annualized), derived from one quarter of data, can easily span ± 1.5. Two days provides far less.
4. **No adverse selection model for maker fills.** Limit orders at the bid fill more often when large sellers are incoming, which is the worst possible time to be long. Ignoring adverse selection systematically inflates maker strategy PnL.

6.5 The 0.75 bps Breakeven is an Optimistic Bound

On Binance, VIP maker rebates can reach -0.01 bps (i.e. the exchange *pays* you for liquidity), which on paper makes the gross alpha sufficient. In practice:

- **Fill rate uncertainty.** Maker orders are not guaranteed to fill. If the fill rate is 60%, the effective cost per intended-trade is higher than the quoted fee tier.
- **Queue position.** In a fast-moving order book, a new limit order joins at the back of the queue. For a mean-reversion trade, where price may start recovering before the order reaches the front, fill quality degrades.
- **Co-location and infrastructure.** Low-latency execution systems (co-location, direct market access, kernel bypass networking) have costs that should be amortized against strategy PnL.

6.6 Two Days is Not a Dataset

The most fundamental limitation is sample size. Two trading days capture a single, relatively quiet market regime (BTC price ranging \$84k–\$86k, a 1.5% window). There is no ability to test:

- How the signal performs in high-volatility regimes (post-news, liquidation cascades, funding spikes).
- Whether the negative lag-1 autocorrelation is a stable feature or a regime-specific artefact.
- Seasonal effects (weekend vs weekday, options expiry days).
- Longer drawdown periods that would expose the fragility of the hysteresis thresholds.

6.7 What Would Make This Credible

For the strategy to be taken seriously in a production context, the following would need to be demonstrated:

1. ≥ 6 **months of tick data** spanning multiple volatility regimes.
2. **Proper walk-forward with parameter optimization**: LASSO/ridge regression on signal weights, or a simple equal-weight ensemble baseline, estimated in-sample and frozen out-of-sample.
3. **Queue simulation**: modeling fill probability as a function of order size, time-in-queue, and market depth.
4. **IC t-statistics with multiple-testing correction** across the 8 signals to assess which ones are genuinely informative.
5. **Intraday volume normalization**: scaling signals by the expected volume in a given hour to avoid regime-mixing.

7. Conclusion

This project demonstrates a complete quantitative research pipeline from raw tick data to cost-aware backtesting for BTCUSDT perpetual futures. The key empirical finding — **negative lag-1 autocorrelation in 1-minute returns** — provides a sound microstructure basis for a mean-reversion signal suite. All price-based signals show positive information coefficient at the 5–10 minute horizon, and the composite signal achieves an IC of approximately 0.10, which is a reasonable starting point for a high-frequency mean-reversion strategy.

The backtesting results indicate **gross alpha that is real but thin**. The strategy is profitable in simulation only below $\sim$0.75 bps in round-trip transaction costs — a threshold achievable with maker limit-order execution on Binance VIP tiers, but not with standard taker fills.

The critical analysis in Section 6 highlights several areas where the simulation is optimistic: signal redundancy between z-score and Bollinger signals, a mis-specified momentum signal (Trade Intensity), inflated Sharpe ratios due to unrealistic fill assumptions, and the fundamental limitation of a two-day dataset.

The work should be viewed as a **research proof-of-concept** rather than a deployable strategy: it identifies a plausible alpha source, validates it with appropriate IC and quintile analysis, and quantifies the execution-cost constraint clearly. Extending the dataset and incorporating realistic execution simulation would be the natural next steps before live deployment.